

Technical Documentation of ZigooCare

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1. Introduction

ZigooCare is a smart urban sanitation management platform designed to improve sewer monitoring, waste collection operations, and recycling center management in Tunisian municipalities.

The system combines traditional management functionalities with intelligent decision-support tools based on data analysis and predictive algorithms. Its objective is to help municipalities prevent sewer blockages, optimize maintenance interventions, improve waste collection logistics, and reduce the risk of urban flooding.

This document presents the technical architecture, software structure, database organization, and implementation details of the ZigooCare application.

2. System Architecture

ZigooCare follows a modular and object-oriented architecture developed in C++ using the Qt framework.

The application is organized into independent modules responsible for specific business domains:

- Employee Management
- Municipality Management
- Sewer Management
- Collection Teams Management
- Recycling Centers Management

Each module communicates with the database layer while remaining isolated from other modules, ensuring maintainability and scalability.

3. Technologies Used

Table 1: Technologies Used

Technology	Purpose
C++17	Core application logic and business rules

Qt Framework	Main application framework for UI and system integration
Qt Widgets	Desktop interface components
Qt Quick / QML	Declarative UI for interactive components
Qt SQL	Database access and integration with Oracle
Qt Charts	Statistical data visualization and dashboards
Qt Network	HTTP/HTTPS communication with external services
Qt Location + Qt Positioning	Map rendering, GPS handling, and geographic operations
Qt WebEngine	Embedded web content
Qt Serial Port	Communication with Arduino hardware via USB serial
Qt Text To Speech	Voice feedback system
Qt Print Support	PDF report generation and printing
Qt JSON Support	Parsing and constructing JSON for APIs and AI communication
Oracle Database	Persistent storage for system data
Oracle SQL Developer	Database design and administration tool
Python	Execution of AI models and predictive analytics
Ollama	Local AI runtime powering the intelligent assistant
Arduino	Hardware interface for sensors and physical interaction
OpenStreetMap	Base map tiles for geographic visualization
Google Maps	Generates itineraries and opens optimized routes in browser via external navigation
External APIs	Real-time data integration

4. Project Structure

The project follows a modular organization to improve readability and maintainability.

```
src/  
├── Arduino/  
├── WasteCollection/  
└── zigooCareAI/
```

4.1. Folder Description

- **Arduino:** Contains the firmware developed for the Arduino-based hardware module. This component manages communication with the desktop application and controls the physical devices connected to the system, such as buttons, sensors, LEDs, LCD displays, and other input/output peripherals.
- **WasteCollection:** Contains the main desktop application developed in C++ using the Qt Framework. This module implements the graphical user interface, business logic, database interactions, data visualization components, mapping features, notification services, and all core functionalities of the waste management system.
- **zigooCareAI:** Contains the artificial intelligence components of the project, including machine learning models, datasets, data preprocessing resources, training scripts, prediction scripts, and other utilities used for data analysis, forecasting, and intelligent decision support within the system.

This modular architecture ensures a clear separation between hardware control, application logic, and AI functionalities, facilitating development, testing, maintenance, and future extensions of the platform.

5. Database Design

5.1. Database Management System

ZigooCare uses an Oracle relational database to store and manage all application data. Database development, administration, and SQL scripting were performed using Oracle SQL Developer. The

database is designed according to relational modeling principles and ensures data integrity through primary keys, foreign keys, and validation constraints.

5.2. Entity-Association Model

The conceptual data model of ZigooCare is represented by the Entity-Association Diagram available in the docs/ directory. The diagram illustrates the main entities of the system and their relationships.

5.3. Relational Model

The logical database structure is represented by the relational model available in the docs/ directory. It describes the database tables, attributes, primary keys, and foreign key relationships used by the application.

5.4. Main Entities

Employee: Stores employee credentials and authorization data.

Municipality: Stores municipality information and geographical coordinates.

Sewer: Stores sewer monitoring information.

Collection Team: Stores intervention team information.

Recycling Center: Stores treatment center information.

6. Authentication and Security

ZigooCare implements several security mechanisms:

6.1. Authentication

Users authenticate using:

- Employee ID + Password

- Fingerprint

6.2. First-Time Account Setup

New users are required to:

- Create a password
- Setup a fingerprint
- Select a security question
- Provide a security answer

6.3. Password Recovery

Users can reset their password by correctly answering their predefined security question.

6.4. Data Protection

The application implements:

- Password hashing
- Input validation
- SQL injection protection
- Role-based access control

7. Functional Modules

The system is organized into several functional modules, each responsible for a specific domain of waste and infrastructure management. These modules encapsulate business logic and ensure separation of concerns across the platform.

7.1. Employee Management Module

This module is responsible for managing the employees of the ministry who have access to the application. It mainly handles authentication-related data and basic administrative operations.

Main responsibilities:

- Employee account management
- User data maintenance
- Role assignment for system access
- Login credential management

7.2. Municipality Management Module

This module provides tools for monitoring and analyzing municipalities and their environmental status. It acts as a global coordination layer for geographic and risk-based information.

Main responsibilities:

- Municipality and region organization
- Geographic visualization and mapping integration
- Environmental risk and resource analysis
- Reporting and decision support

7.3. Sewer Management Module

This module focuses on sewer network monitoring and maintenance prediction. It supports preventive maintenance strategies through data analysis and forecasting.

Main responsibilities:

- Sewer network monitoring
- Blockage and degradation analysis
- Predictive maintenance support
- Optimization of intervention planning

7.4. Collection Teams Module

This module manages intervention teams responsible for waste collection and field operations. It ensures efficient workforce allocation and performance tracking.

Main responsibilities:

- Team organization and assignment
- Workload distribution and balancing
- Performance evaluation and monitoring
- Operational efficiency improvement

7.5. Recycling Centers Module

This module manages recycling and waste treatment centers, focusing on capacity control and operational efficiency.

Main responsibilities:

- Facility monitoring and management
- Capacity and saturation control
- Operational reporting and analytics
- Notification and alert systems

8. Artificial Intelligence Components

The system integrates two lightweight AI-based features that enhance decision support and user interaction. These components are not standalone machine learning models, but rather intelligent modules that provide automated analysis and assistance.

8.1. Environmental AI (Sustainability Assessment Module)

This component evaluates a selected sewer system from an environmental perspective. It analyzes how the sewer's condition and associated data align with sustainability principles and the United Nations Sustainable Development Goals (SDGs).

Main functions:

- Assessment of environmental impact of a sewer

- Evaluation of sustainability alignment (SDG-oriented analysis)
- Generation of an interpretative score or feedback based on collected parameters

The objective is to support environmentally conscious decision-making in infrastructure management.

8.2. ZigooBot (Voice Assistant)

ZigooBot is a voice-based assistant integrated into the application to improve user interaction and accessibility. It responds to spoken queries and provides contextual information about the system.

Main functions:

- Answering general questions about the project
- Providing explanations of system features and modules
- Delivering specific information on demand via voice interaction

The objective is to facilitate user guidance and improve the accessibility of system information.

9. Reporting and Analytics

The system provides decision-support tools through:

- Statistical dashboards
- Pie charts
- Bar charts
- Capacity monitoring indicators
- Collection team ranking systems
- Saturation alerts

Export formats include:

- Excel
- PDF

10. Notification System

The platform automatically sends email notifications whenever a recycling center receives a new waste assignment.

Emails include:

- Assignment details
- Daily processing totals
- Operational information

Administrators can customize email templates through the Email Settings module.

11. Software Design Diagrams

Table 2: Software Design Diagrams

Diagram	Purpose
Use Case Diagram	Functional view of the system
Entity-Association Diagram	Conceptual database design
Relational Model	Logical database structure

12. Hardware Integration (Arduino-Based System)

The system integrates an Arduino-based hardware component to extend interaction beyond the software interface. This module acts as a physical input/output interface connected to the application.

12.1. Main features

Real-time communication between the Arduino and the application via serial USB

Input acquisition through physical components, including:

- Push buttons used to simulate waste events
- Water level sensor for environmental monitoring
- Rain sensor for detecting weather conditions
- Fingerprint sensor for user identification

Output feedback through:

- LCD display for system information and status messages
- Buzzer for alerts and notifications

This integration enhances user interaction by providing a tangible interface for monitoring system conditions and triggering or visualizing events in real time.

13. Maintainability and Extensibility

The application was designed according to modular software engineering principles.

Advantages include:

- Easy addition of new modules
- Simplified maintenance
- Improved code readability
- Reusable components
- Independent testing of modules

The architecture allows future integration of:

- IoT sewer sensors
- Mobile applications
- Advanced AI models
- Real-time monitoring services

14. Conclusion

ZigooCare provides a complete smart sanitation management solution combining operational management, predictive analytics, and decision-support tools.

Its modular architecture, integration of artificial intelligence, and focus on preventive maintenance make it a scalable platform capable of supporting modern urban sanitation management and sustainable city development.